Society V1
Anatomy of the 42

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The 42 conceptual variables that compose Society V1 represent abstractions of the fundamental forces operating within society. These concepts (e.g., DIVERSITY, POVERTY, MENTAL HEALTH, etc.) are mathematically defined by an influence matrix, a collection of output edges with weighted polarity expressed as normalized values between -1 and 1. Each variable is materially tethered to a physical slider on the Society V1 controller board, allowing users to interact directly with the model.

Changing the value of a variable by moving its corresponding slider adjusts the directional intensity of the influence it exerts on others connected to it. No change occurs in isolation; rather, it incites conditional responses along associated pathways, dynamically reshaping the collective state of the system.

To illustrate: If you change the value of LAW ENFORCEMENT, depending on the current vector state of the system, CRIME will change its state by an amount determined by the weighted edge connecting the two variables. When CRIME settles into its new position, its own network of influences will trigger, affecting downstream variables like DEATH RATE and PUBLIC TRUST.

The mechanism may appear straightforward, but it naturally raises important questions. How are the weights assigned? What justifications or rationale support the exact values of these weights? Are they speculative? Is it arbitrary? How can a specific value be deduced from empirical research when the findings themselves often present convoluted observations or abstract outcomes inferred by researchers?

This document outlines how the values for these concepts are informed by and inferred from aggregated research findings, comparative data, and disciplined observational reasoning.

For readers interested in the broader overview of the system, see *Society V1: Whitepaper*. For the technical implementation, including the influence engine and mathematical foundations, see *Computational & Mathematical Architecture (CMA)*.

It should be underscored that these values, along with the mathematics directing them, are ultimately transcribed by a single mortal mind[1] attempting, through large bodies of research and literature, to painstakingly quantify abstractions far more complex than any model can fully capture. The act of bounding these broad abstractions into discrete values inevitably reduces their individual richness; however, at scale, when enough are structured together, this reduction becomes the very mechanism that makes modeling society possible.

The 42

1. Addiction
2. Agriculture
3. Authoritarianism
4. Birthrate
5. Corruption
6. Cost of Living
7. Crime
8. Death Rate
9. Discrimination
10. Diversity
11. Education Access
12. Education Cost
13. Education Quality
14. Employment Rate
15. Energy Production
16. Equality
17. Family Values
18. Freedom of Speech
19. Gun Rights
20. Healthcare Access
21. Healthcare Cost
22. Healthcare Quality
23. Homelessness
24. Immigration
25. Infrastructure
26. Innovation
27. Law Enforcement
28. Liberty
29. Market Freedom
30. Mental Health
31. Military
32. Minimum Wage
33. Pollution
34. Poverty
35. Public Trust
36. Purpose/Meaning
37. Spirituality/God
38. Sickness/Disease
39. Taxes
40. Technology
41. Transportation
42. Welfare Support

[1] Currently developed by a single human modeler, though additional minds would likely improve both the model and the modeler.

➠ Interpretation Framework

Before jumping down the rabbit hole, I feel it would be wise to provide the reader with context for how each variable is interpreted and structured. To avoid confusion, all variable descriptions in this document will follow a consistent formatting framework. Each section will include a detailed description, methodological scope, and acknowledgment of any known challenges or disputes within the academic literature.

Ontology

Society does not consist of cleanly separable components with absolute borders or dividing lines. In reality, many social phenomena overlap, interact, repel, and co-evolve in extremely complex ways. Because of this, clear conceptual distinctions must be intentionally drawn in order to model them.

For the purposes of Society V1, each variable is assigned abstract boundaries. In the sections that follow, these abstractions are outlined and systematized to minimize conceptual overlap while preserving the ability to model emergent system behavior with reasonable fidelity.

Inclusionary & Exclusionary Scopes

Inclusionary and exclusionary lists define the conceptual perimeter of each variable. These boundaries clarify what is intentionally considered and what is not included within the boundaries of each conceptual variable, ensuring shared interpretation and internal consistency.

This document aims to answer practical questions such as:

- What does each variable (or their numerical values) actually mean?
- What am I changing when I move a variable's value up or down?
- How are these concepts measured in the real world?
- How are specific influence values determined when research findings vary across studies?

Research Metrics & Methodology[2]

All sections will contain the relevant quantitative metrics, secondary indicators, and research approaches corresponding to the verifiable data used to ground each variable's influence weight to the real world.

Model parameters are derived through a structured synthesis of:

- Established empirical research
- Cross-national and national statistical datasets
- Peer-reviewed literature
- AI tools[3]
- Comparative historical analysis

Path-Dependent Influence Weights

Variables are defined broadly, but their output pathways (influence edges) are formulated contextually. Each variable influences others differently, depending on the pathway through which the interaction occurs. In other words, every variable casts different shadows depending on where it lands. Accordingly, each influence edge weight is inferred from the empirical metrics most directly associated with the connection being modeled.

For example, ADDICTION is not simply a singular influence weight applied uniformly across different receptor variables (such as CRIME or LIBERTY). Instead, it functions as a matrix of path-specific influence weights that exert discretely different intensities across its downstream relationships. ADDICTION'S influence on DEATH RATE is evaluated through overdose and attributable mortality, while its influence on MENTAL HEALTH is interpreted through the prevalence of dual-diagnosis, surveys, and hospitalization rates.

In this way, the model maintains empirical grounding while avoiding the false assumption that a complex societal variable can be represented by a single uniform measurement.

[2] To find Society V1's citations, datasets, literature, and research sources see: RESEARCH REPOSITORY

[3] Including platforms such as Consensus AI for literature aggregation and comparative analysis.

1 Addiction

ADDICTION is the prevalence and severity of compulsive dependency behaviors within a population. This category encompasses addiction to psychoactive substances (e.g., alcohol, opioids, stimulants, tobacco, etc.) as well as compulsive behavioral dependencies independent of chemical exposure (e.g., gambling, sex, pornography, digital dependency, etc.).

This variable is modeled by placing emphasis on the relevant mechanisms of addiction that propagate into broader societal effects, at the population level, rather than individual experiences at the psychological level.

Independent of the specific substance or behavior involved, ADDICTION is conceptualized along two primary dimensions: *Physiological Dependency* and *Psychological Compulsion*

» ADDICTION includes:

- Pervasiveness of active addiction
- Severity and chronicity
- Recovery and relapse dynamics
- Economic, judicial, and social burdens
- Social and institutional degradation

» ADDICTION excludes:

- General mental illnesses
- Occasional recreational use
- Addiction-related mortality
- Non-compulsive binge behavior
- Criminality

When the value of ADDICTION increases, it represents a greater prevalence and severity of dependency within the population; conversely, when it decreases, those burdens recede.

In Society V1, ADDICTION functions as both a public health burden and a social destabilizer, influencing productivity, mental health, family stability, crime rates, and broader societal resilience. ADDICTION is supported by a robust empirical foundation, allowing it to be modeled with a relatively high degree of confidence.

Research Metrics & Methodology

Primary	Secondary
Substance Use Disorder (SUD) prevalence	Self-reported dependency surveys
Overdose rates	Drug-related economic burden estimates
Treatment enrollment rates	Workplace impairment statistics
Relapse rates	Dual Diagnosis Prevalence (psychiatric comorbidity)
Hospitalization Statistics	

Data Challenges

- Under-reporting
- Inconsistencies in cross-national reporting
- Measurement gaps
- Cultural normalization differences
- Social stigma; dampening treatment and diagnosis visibility
- Misdiagnosis; also including Over- and Under-diagnosis

These challenges mean that addiction data should be interpreted with caution. Under-reporting and social stigma make addiction harder to detect in surveys, while inconsistent reporting standards across countries make direct comparisons unreliable. Because of this, most available statistics likely underestimate the true scope of addiction.

Known Confounds & Academic Disputes

- Criminalization vs public health framing
- Direction of Socioeconomic causality: (Poverty → Addiction) vs. (Addiction → Poverty)
- Measurement inconsistency in clinical and survey definitions

These disputes reflect unresolved questions regarding causality and framing. For example, whether addiction is primarily a public health issue or a criminal justice issue influences data collection practices and policy response, which in turn affect reported prevalence and outcome measures.

Addiction Edges	Weight	Mechanism Summary
Agriculture	-0.08	Weakens productivity and increases workplace accidents
Birthrate	-0.16	Reduces child birth, though at higher levels the curve bends slightly back upward as unplanned pregnancies become more common
Cost of Living	+0.15	Addiction increases economic burden through health care costs, productivity loss, crime-related expenses, and housing instability, which collectively add pressure on the cost of living
Crime	+0.3	Addiction increases criminal activity through intoxication effects, economic compulsion to finance substance use, and broader social destabilization.
Death Rate	+0.31	xxxxxxxx xxxxxxx xxx
Education: Access	-0.07	xxxxxxxx xxxxxxx xxx
Education: Quality	-0.13	xxxxxxxx xxxxxxx x
Employment	-0.18	xxxxxxxx xxxxxx
Equality	-0.06	xxxxxxxx xxxxxxx
Family Values	-0.21	xxxxxxxx xxxxxxx xx
Gun Rights	-0.12	xxxxxxxx xxxxxxx xxx
HC Access	-0.13	xxxxxxxx xxxxxxx xxx
HC Cost	+0.17	xxxxxxxx xxxxxxx x
HC Quality	-0.14	xxxxxxxx xxxxx
Homelessness	+0.3	xxxxxxxx xxxx
Immigration	-0.07	xxxxxxxx xx
Innovation	-0.2	xxxxxxxx xxxx
Law Enforcement	+0.25	xxxxxxx
Liberty	-0.21	xxxxxxxx xxxx
Market Freedom	-0.11	xxxxxxxx
Mental Health	-0.35	xxxxxxxx x
Military	-0.16	xxxxxxxx
Poverty	+0.33	more filler text because dingleboobs f
Public Trust	-0.12	xxxxxxxx
Purpose/Meaning	-0.38	xxxxxxxx xx
Sickness/Disease	0.31	xxxxxxxx x
Spirituality/God	-0.17	xxxxxxxx
Welfare	+0.22	xxxxxxxxxxx

2 Agriculture

AGRICULTURE is much more than the total seasonal harvest projected into a commodity chart that a hedge-fund manager can present in a boardroom or an economic conference; it is the quiet mother of civilization itself. Rather than a simple capacity, it is a dense system variable: fields and livestock, management and labor, machines and ecological sustainability, cross-sector coordination, soil, logistics, knowledge, capital dynamics, technological adoption, and the living choreography that keeps agricultural products flowing through store shelves and onto the tables of hundreds of millions of families around the world.

In other words, it asks not only how much the system produces, but how well it works, how stable it is, and whether it can keep working under pressure.

» AGRICULTURE includes:

- Productive capacity of crops and livestock
- Quality of labor within agricultural systems
  - Agricultural workforce availability
  - Health, skill level, and productivity
- Mechanization, tools, and technological integration
- Infrastructure and logistical reliability
- Environmental Sustainability and soil vitality
- Community-level agricultural cohesion and knowledge transmission

» AGRICULTURE exludes:

- Broader industrial output
- Climate dynamics
- National wealth outside agricultural systems
- Agricultural regulations, laws, or policies

Agriculture is a foundational stabilizer in the model because it is a foundational stabilizer in reality. When agricultural systems are strong, food security, economic stability, rural communities, and public health tend to stand upright with it. When agriculture weakens, fragility spreads outward in quiet but compounding ways.

Research Metrics & Methodology

Primary	Secondary
Crop yield stability	Input efficiency (fertilizer, energy, water, etc.)
Livestock productivity	Income stability
Food affordability pressure	Post-harvest loss rates
Agricultural labor productivity	Soil health and land degradation
Production-storage-distribution continuity	Import dependence and commodity-shock exposure

Data Challenges

- Subsistence and informal production are often under-counted
- Cross-country reporting standards are inconsistent
- Climate shocks can obscure structural agricultural performance
- Policy and subsidy regimes can distort market-facing indicators
- Many effects are lagged, so short windows can misrepresent direction and strength

These limitations mean Agriculture should be interpreted as a systems variable, not a single-output statistic. Short-term output can improve while structural resilience weakens, or vice versa.

Known Confounds & Academic Disputes

- High-output industrial models vs long-run ecological sustainability
- Scale efficiency vs resilience of distributed/local food systems
- Direction of causality in development loops (Poverty:low → Agriculture vs. Agriculture → prosperity:low)
- Institutional quality and infrastructure acting as hidden mediators in social outcome pathways

Because research does not always agree, the influence values in Society V1 are not copied from any single study. They are informed by broader mechanisms, consistent directional signals, and the clearest patterns that appear across multiple sources. As better evidence emerges, the values can be adjusted and refined.

2.1 Birthrate

description here

2.2 Corruption

description here

2.3 Cost of Living

description here

2.4 Crime

description here

2.5 Death Rate

description here

2.6 Discrimination

description here

2.7 Diversity

description here

2.8 Education Access

description here

2.9 Education Cost

a

2.10 Education Quality

a

2.11 Employment

a

2.12 Energy Production

a

2.13 Equality

a

2.14 Family Values

a

2.15 Freedom of Speech

a

2.16 Gun Rights

a

2.17 Healthcare Access

a

2.18 Healthcare Cost

a

2.19 Healthcare Quality

a

2.20 Homelessness

a

2.21 Immigration

a

2.22 Infrastructure

a

2.23 Innovation

a

2.24 Law Enforcement

a

2.25 Liberty

a

2.26 Market Freedom

a

2.27 Mental Health

a

2.28 Military

a

2.29 Minimum Wage

a

2.30 Pollution

a

2.31 Poverty

a

2.32 Public Trust

a

2.33 Purpose/Meaning

a

2.34 Sickness/Disease

a

2.35 Spirituality/God

a

2.36 Taxes

a

2.37 Technology

a

2.38 Transportation

a

2.39 Welfare Support

a

3 Template Sheet (replace with variable name)

NAME's description

- NAME includes:
  - 1
  - 2
  - 3
  - 4
  - ...
- What this variable excludes:
  - 1
  - 2
  - 3
  - 4
  - ...

Model description for NAME

Research Metrics & Methodology

Primary	Secondary
p1	s1
p2	s2
p3	s3
p4	s4
p5	s5

Data Challenges:

1
2
3
4
5
6

Known Confounds & Literature Disputes

1
2
3
4
5